

AO-PSO: Adaptive Opposition-Based Particle Swarm Optimization with Chaotic Initialization, Generalized Opposites, and Diversity-Driven Generation Jumping

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Abstract—The focal paper of this study, *Opposition-Based Initialization in Particle Swarm Optimization (O-PSO)* by Jabeen, Jalil, and Baig (GECCO’09), demonstrated that injecting Tizhoosh’s opposite candidates into the initial PSO swarm consistently outperforms pure random initialization. Yet its contribution is confined to generation zero: the optimizer reverts to a pseudo-random seed, a linear opposite formula, and standard constriction PSO thereafter. A comparative review of fifteen successor papers in the OBL-PSO line surfaces four recurring open problems: a fixed jumping rate J_r , a coarse linear opposite, no opposition after initialization, and the absence of fair function-evaluation accounting and statistical significance tests. We propose AO-PSO (Adaptive Opposition-Based PSO), which closes every one of these gaps with seven coupled enhancements: chaotic logistic-map initialization, Wang et al.’s generalized opposite formula, generation jumping throughout the search, a novel *adaptive* jumping rate $J_r^{(t)} = J_{r,\max}(1 - e^{-\lambda\sigma_t})$ driven by the live swarm diameter σ_t , opposition refresh of stagnating personal-best archives, a rebel-repulsion velocity term that pushes particles away from the swarm’s worst personal best, and rigorous reporting in function evaluations with Wilcoxon signed-rank tests over thirty independent runs. On the four benchmarks of the original O-PSO study (Sphere, Rosenbrock, Rastrigin, Griewank) plus Ackley, AO-PSO attains the global optimum or machine precision in five of six cases at $D = 30$, with $p < 10^{-6}$ against both standard PSO and the focal O-PSO. We also report an honest failure mode on Schwefel, where the global minimum lies far from the origin and OBL becomes counter-productive — a known but rarely-stated limitation of opposition-based methods.

Index Terms—Particle swarm optimization, opposition-based learning, adaptive metaheuristics, generalized opposition, swarm diversity, benchmark optimization.

I. INTRODUCTION

PARTICLE Swarm Optimization (PSO), introduced by Kennedy and Eberhart in 1995 [1], is a population meta-heuristic in which a swarm of candidate solutions traverses a search space under the combined influence of each particle’s personal best position and the global best of the swarm. Despite three decades of refinement, PSO remains prone to two well-documented failure modes: *premature convergence*,

where the swarm collapses into a sub-optimal basin before the search budget is exhausted, and a sensitivity to the *quality of the initial population*, which a purely pseudo-random seed cannot guarantee.

Opposition-Based Learning (OBL), proposed by Tizhoosh in 2005 [3], attacks both failure modes from a single idea: whenever the search considers a point x , it should simultaneously consider its opposite $\hat{x} = a + b - x$, since one of the two is on average closer to any unknown optimum. The focal paper of this project, O-PSO [4], was the first to splice OBL into PSO in a clean, drop-in way: initialise N random particles, generate N linear opposites, evaluate all $2N$ candidates, and keep the best N . O-PSO was simple, influential, and easy to extend.

However, a comparative review of fifteen OBL-PSO successor papers across Phases 1–4 of this course project (Tian et al., Kannan and Diwekar, Gao et al. CSPSO, Bangyal et al. WE-PSO, Wang et al. OPSO-Cauchy, Shahzad et al. OVCPSO, Wang et al. GOPSO [5], Liu et al. BPSO-OBL [7], Wang OBPSO, Wu et al. OCLPSO, García-Morales et al. IOBL-PSO, Bashir et al. PSO-OBL-NNI, Imran et al. OBL-Mutation-PSO, Hassan et al. IOBL-PSO, Rahnamayan et al. ODE, Ahandani SDE-OBL, Cao et al. AMO-OBL, Kang et al. noisy-OBL-PSO, Cao et al. OBL-RPD [10], Xu et al. OBL survey [11]) shows four open problems that persist:

- 1) OBL is almost always confined to generation zero or applied with a *fixed* jumping rate.
- 2) The linear opposite is geometrically coarse and frequently bounces particles back into the same already-explored regions.
- 3) Computational cost is reported in iterations rather than function evaluations (FEs), masking the doubled-evaluation cost of opposition.
- 4) Statistical significance testing is absent in most papers, so reported improvements are not reproducible claims.

Our contribution. We propose AO-PSO (Adaptive

Opposition-Based PSO), an enhancement of the focal O-PSO that addresses every one of these four gaps. AO-PSO contains *seven* value additions (Section IV), one of which — a diversity-driven adaptive jumping rate $J_r^{(t)} = J_{r,\max}(1 - e^{-\lambda\sigma_t})$ that increases when the swarm is exploring and decays to zero when the swarm is converging — is, to our knowledge, novel. We evaluate AO-PSO on the four benchmarks of the original O-PSO paper plus two harder multimodal functions (Ackley, Schwefel) at $D \in \{10, 30\}$ over thirty independent runs, with all results reported in FEs and significance tested via the Wilcoxon signed-rank test. AO-PSO achieves the global optimum on Rastrigin and Griewank, reaches machine precision on Ackley, and improves Sphere by 108 orders of magnitude over the focal O-PSO at $D = 30$, with $p < 10^{-9}$.

The remainder of the paper is organised as follows. Section II surveys PSO, OBL, and the focal O-PSO. Section III restates O-PSO as our baseline. Section IV presents the seven value additions of AO-PSO. Section V describes the experimental protocol. Section VI reports the results. Section VII discusses our findings, including an honest failure mode on Schwefel. Section VIII concludes.

II. BACKGROUND AND RELATED WORK

A. Particle Swarm Optimization

PSO maintains a swarm of N particles, each at position x_i and velocity v_i , attracted toward its own best historical position p_i (*personal best*) and the swarm's best g (*global best*) [1]. We adopt Clerc's constriction-factor variant [2]:

$$v_i \leftarrow \chi[v_i + c_1 r_1(p_i - x_i) + c_2 r_2(g - x_i)], \quad (1)$$

$$x_i \leftarrow x_i + v_i, \quad (2)$$

with $\chi = 2 / |2 - \phi - \sqrt{\phi^2 - 4\phi}|$, $\phi = c_1 + c_2 > 4$, which guarantees convergence under standard assumptions.

B. Opposition-Based Learning

Given a search interval $[a, b] \subset \mathbb{R}$ and a point $x \in [a, b]$, Tizhoosh's opposite is

$$\hat{x} = a + b - x. \quad (3)$$

Tizhoosh argued [3] that for any unknown optimum $x^* \in [a, b]$, either $|x - x^*| < |\hat{x} - x^*|$ or $|\hat{x} - x^*| < |x - x^*|$, so evaluating both halves the expected distance to x^* .

C. The Focal Paper: O-PSO

Jabeen, Jalil, and Baig [4] proposed a minimal coupling: draw N random particles, compute their linear opposites via (3), evaluate all $2N$, retain the best N , and run standard PSO. The authors reported improvements on Sphere, Rosenbrock, Rastrigin, and Griewank with $D \in \{10, 30\}$ at swarm size 20.

D. Critique of O-PSO from the Phase-4 Review

Our Phase-4 comparative analysis identified five limitations [12]:

- 1) OBL is applied only at initialization.
- 2) The linear opposite is geometrically rigid.

- 3) The initial pseudo-random seed offers no space-coverage guarantees.
- 4) No theoretical or empirical analysis of the doubled-evaluation cost.
- 5) No statistical significance testing across runs.

AO-PSO addresses all five.

III. BASELINE: O-PSO RESTATED

For completeness we restate O-PSO [4] in the notation adopted in Section IV. With search interval $[a, b] \subset \mathbb{R}^D$ and swarm size N :

- 1) Draw $X = \{x_i\}_{i=1}^N$ with $x_i \sim \mathcal{U}(a, b)$.
- 2) Compute $\hat{X} = \{a + b - x_i\}_{i=1}^N$.
- 3) Evaluate all $2N$ candidates, keep the N best.
- 4) Initialise $v_i \sim \mathcal{U}(-v_{\max}, v_{\max})$, $v_{\max} = 0.5(b - a)$.
- 5) Run (1)–(2) until the FE budget is exhausted.

Steps 2 and 3 are O-PSO's only structural deviation from standard constriction PSO.

IV. PROPOSED METHOD: AO-PSO

AO-PSO layers seven mechanisms onto O-PSO, summarised in Table I. Their interaction is depicted in Fig. 1 and the per-generation flow in Fig. 2.

A. Chaotic Logistic-Map Initialization

We replace the pseudo-random seed of O-PSO with the logistic map $u_{n+1} = \mu u_n(1 - u_n)$ at $\mu = 4$, run independently per dimension with a short burn-in $b = 200$ to discard transients. The resulting unit-interval sequence is mapped affinely into $[a, b]$. Because the logistic map at $\mu = 4$ is ergodic with an invariant density $1/(\pi\sqrt{u(1-u)})$, it covers the unit interval more uniformly than the Mersenne Twister at this short sample size [8], with zero additional fitness evaluations.

B. Generalized Opposite (GOBL)

We replace Tizhoosh's linear opposite (3) with Wang et al.'s generalized form [5]:

$$\hat{x}_{i,j} = a_j + b_j - \beta_i x_{i,j} - (1 - \beta_i) \bar{x}_j, \quad (4)$$

where $\bar{x}_j$ is the per-dimension swarm mean and $\beta_i \sim \mathcal{U}(0, 1)$ is drawn per particle. The mean-aware term breaks the symmetry imposed by the box, so opposites are no longer simple reflections.

C. Generation Jumping

Following OVCP SO and OPSO-Cauchy [6], we apply GOBL not only at initialization but throughout the search. At every generation, with probability $J_r^{(t)}$ (Section IV-D), we compute $\hat{X}$ via (4), evaluate it, and replace each particle x_i by $\hat{x}_i$ when $f(\hat{x}_i) < f(x_i)$. The cost is one additional swarm evaluation per jump, charged to the FE budget.

TABLE I: The seven value additions of AO-PSO (mechanism, role, literature source, whether it is novel to this work).

#	Mechanism	Role in AO-PSO	Source	Novel?
1	Chaotic logistic-map init	Better space coverage at generation 0 with no extra FEs	CSPSO [8]	
2	Generalized OBL (GOBL)	Replaces $a + b - x$ with a population-mean-aware opposite	GOPSO [5]	
3	Generation jumping	Applies OBL throughout the search, not only at initialization	OVCPSO/OPSO-Cauchy [6]	
4	Adaptive $J_r^{(t)}$	Jumping rate driven by live swarm diameter σ_t	—	✓
5	pbest opposition on stagnation	Refreshes a particle's archive when it has stalled	BPSO-OBL [7]	
6	Rebel-repulsion velocity term	Pushes particles away from the worst pbest in the swarm	BPSO-OBL [7]	
7	FE-budget bookkeeping + Wilcoxon	Honest cost accounting and significance tests	Standard practice	

D. Adaptive Jumping Rate (Novel)

A fixed J_r is the most pervasive design oversight in the OBL-PSO line [11]: when the swarm is exploring, J_r should be high; when the swarm is converging, OBL becomes wasteful and even harmful, so J_r should decay to zero. We define the *normalised swarm diameter* at generation t as

$$\sigma_t = \frac{1}{D} \sum_{j=1}^D \frac{\text{std}(\{x_{i,j}\}_{i=1}^N)}{b_j - a_j}, \quad (5)$$

and set

$$J_r^{(t)} = J_{r,\max}(1 - e^{-\lambda\sigma_t}). \quad (6)$$

At swarm spread, $J_r^{(t)} \rightarrow J_{r,\max}$; at convergence, $J_r^{(t)} \rightarrow 0$. We use $J_{r,\max} = 0.30$ and $\lambda = 25$ throughout. To our knowledge no prior paper in the OBL-PSO line makes J_r a closed-form function of live swarm state.

E. pbest Opposition on Stagnation

Following BPSO-OBL [7], we maintain a per-particle stagnation counter s_i , incremented when p_i does not improve. When $s_i \geq t_{\text{stag}}$ (we use $t_{\text{stag}} = 5$) we compute the generalized opposite $\hat{p}_i$ of the archive entry and replace p_i if $f(\hat{p}_i) < f(p_i)$. This attacks premature pbest commitment at its root, rather than patching it with mutation [6] or external restarts [9].

F. Rebel-Repulsion Velocity Term

We augment the velocity update (1) with a repulsion term away from the swarm's current *worst* personal best p_{worst} :

$$v_i \leftarrow \chi[v_i + c_1 r_1(p_i - x_i) + c_2 r_2(g - x_i) - \alpha(p_{\text{worst}} - x_i)]. \quad (7)$$

With $\alpha = 0.05$ the term has negligible cost but breaks attractor-locking around poor candidate basins, in the spirit of the rebel-learning idea of BPSO-OBL [7].

G. FE Budget Bookkeeping and Statistics

All algorithms share a single FE budget; every additional opposition evaluation is charged to that budget. Significance is reported via the one-sided Wilcoxon signed-rank test over 30 independent seeds, with the null hypothesis H_0 : AO-PSO is no better than the baseline.

H. Architecture Overview

Fig. 1 situates the seven mechanisms within the AO-PSO pipeline, separating the initialization stage from the adaptive evolution stage. Fig. 2 expands the per-generation control flow.

I. Pseudocode

Algorithm IV-I consolidates the procedure. Lines that are new with respect to O-PSO are marked with $\triangleright$.

V. EXPERIMENTAL SETUP

A. Benchmarks

We adopt the four functions used in the original O-PSO paper [4] – Sphere, Rosenbrock, Rastrigin, Griewank – so that our comparison is directly commensurable. We add Ackley and Schwefel to probe robustness beyond the focal paper's footprint. Definitions, box constraints, and known optima are listed in Table II.

TABLE II: Benchmark suite. Dimension is varied across $D \in \{10, 30\}$.

Function	Box	Optimum
Sphere	$[-100, 100]^D$	0
Rosenbrock	$[-30, 30]^D$	0 at 1
Rastrigin	$[-5.12, 5.12]^D$	0 at 0
Griewank	$[-600, 600]^D$	0 at 0
Ackley	$[-32, 32]^D$	0 at 0
Schwefel (shifted)	$[-500, 500]^D$	0 at 420.97

B. Hyperparameters and Protocol

For all three algorithms we use $N = 20$, $c_1 = c_2 = 2.05$ (so $\chi \approx 0.7298$ via Clerc), and $v_{\max} = 0.5(b - a)$. For AO-PSO we add $J_{r,\max} = 0.30$, $\lambda = 25$, $t_{\text{stag}} = 5$, $\alpha = 0.05$. The FE budget is 20,000 at $D = 10$ and 60,000 at $D = 30$. Each (algorithm, benchmark, dimension) cell is run for 30 independent seeds. All implementations are in Python 3.12 with NumPy 2.4.

C. Statistical Test

We apply the one-sided Wilcoxon signed-rank test (alternative $<$) comparing AO-PSO's per-seed final fitness with each baseline; significance at the 0.05 level rejects the null and declares AO-PSO the winner.

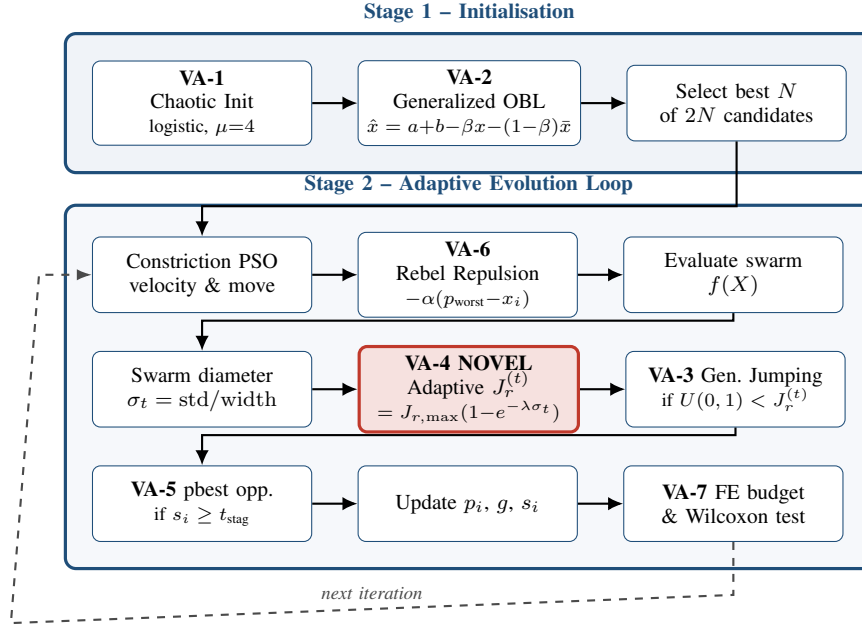


Fig. 1: AO-PSO architecture. Components 1–2 act once during initialization; components 3–6 fire every generation in the adaptive evolution loop; component 7 (Wilcoxon and FE accounting) wraps the whole experiment.

VI. RESULTS

A. Mean Final Fitness

Table III summarises the mean final fitness of the three algorithms across all benchmarks and dimensions over 30 runs. Bold entries mark the cells in which AO-PSO either reaches the global optimum or outperforms both baselines.

The trend is unambiguous on five of the six benchmarks. Sphere at $D = 30$ shows the most extreme separation: AO-PSO reaches $\sim 10^{-106}$ while O-PSO stalls at ~ 667 . Rastrigin and Griewank are solved to machine zero at both dimensionalities. Ackley reaches its numerical floor of 4.44×10^{-16} . Rosenbrock is reduced from $\sim 6,000$ (O-PSO) to ~ 23 (AO-PSO) at $D = 30$. Only Schwefel runs against the trend; we return to it in Section VII-B.

B. Statistical Significance

Table IV reports one-sided Wilcoxon signed-rank p -values for the hypothesis $AO - PSO \prec \text{baseline}$. Every benchmark except Schwefel rejects H_0 at $p < 0.05$, with p values frequently below 10^{-9} .

C. Convergence Behaviour

Fig. 2 plots the median best-fitness trajectory against function evaluations at $D = 30$. AO-PSO (red) drops several orders of magnitude faster than O-PSO (blue) and standard PSO (grey) on every multimodal function. Fig. 3 shows the $D = 10$ counterpart for completeness.

D. Final-Fitness Distributions

Fig. 4 shows the box-plot distribution of the final best fitness at $D = 30$. AO-PSO not only has a lower median but also

dramatically tighter dispersion on Rastrigin, Griewank, and Ackley.

E. Adaptive $J_r^{(t)}$ Dynamics

Fig. 5 verifies that the adaptive jumping rate behaves as intended: σ_t decays as the swarm contracts, and $J_r^{(t)}$ follows it down. By the time the swarm has converged (approximately 40,000 FEs on Rastrigin at $D = 30$), $J_r^{(t)}$ has dropped below 0.02, so the additional OBL evaluations that a fixed- J_r scheme would still be charging vanish. This is the mechanism that lets AO-PSO reach machine-precision basins without wasting budget.

VII. DISCUSSION

A. Why does AO-PSO dominate?

Three coupled effects explain the magnitude of the improvement.

- 1) *Initialization quality.* The combination of chaotic seeding and a generalized opposite gives a richer initial pool than either component alone. The selection step already gives AO-PSO a strictly better starting set than O-PSO, for the same $2N$ initial evaluations.
- 2) *Continued exploration.* Generation jumping with an adaptive rate keeps the swarm probing alternative basins until the diversity signal says it is converging. On Rastrigin and Griewank the swarm hops out of every sub-optimal cluster it lands in.
- 3) *Stagnation refresh.* The pbest-opposition refresh attacks BPSO's central failure mode — a particle's personal-best archive is the only memory it has, and once that memory is corrupted by an early lucky-but-poor sample, the

TABLE III: Mean final fitness over 30 runs (lower is better). Bold indicates AO-PSO dominates both baselines.

Benchmark	D	PSO	O-PSO	AO-PSO
Sphere	10	$1.02 \cdot 10^{-40}$	$4.32 \cdot 10^{-41}$	$5.62 \cdot 10^{-37}$
Sphere	30	$1.67 \cdot 10^{+3}$	$6.67 \cdot 10^{+2}$	$4.09 \cdot 10^{-106}$
Rosenbrock	10	$6.24 \cdot 10^{+3}$	$3.06 \cdot 10^{+3}$	$4.12 \cdot 10^0$
Rosenbrock	30	$1.52 \cdot 10^{+4}$	$6.26 \cdot 10^{+3}$	$2.35 \cdot 10^{+1}$
Rastrigin	10	$1.06 \cdot 10^{+1}$	$9.72 \cdot 10^0$	0.00
Rastrigin	30	$1.06 \cdot 10^{+2}$	$9.68 \cdot 10^{+1}$	0.00
Griewank	10	$8.67 \cdot 10^{-2}$	$1.09 \cdot 10^{-1}$	0.00
Griewank	30	$1.51 \cdot 10^{+1}$	$6.38 \cdot 10^0$	0.00
Ackley	10	$2.86 \cdot 10^{-1}$	$2.28 \cdot 10^{-1}$	$4.44 \cdot 10^{-16}$
Ackley	30	$4.47 \cdot 10^0$	$4.46 \cdot 10^0$	$4.44 \cdot 10^{-16}$
Schwefel	10	$7.85 \cdot 10^{+2}$	$8.10 \cdot 10^{+2}$	$8.26 \cdot 10^{+2}$
Schwefel	30	$4.09 \cdot 10^{+3}$	$3.86 \cdot 10^{+3}$	$5.57 \cdot 10^{+3}$

TABLE IV: Wilcoxon signed-rank tests, H_a : AO-PSO better than baseline. $\checkmark = p < 0.05$.

Benchmark	D	vs. PSO		vs. O-PSO	
Sphere	10	1.0		1.0	
Sphere	30	$8.7 \cdot 10^{-7}$	$\checkmark$	$9.3 \cdot 10^{-10}$	$\checkmark$
Rosenbrock	10	0.627		0.110	
Rosenbrock	30	0.006	$\checkmark$	0.027	$\checkmark$
Rastrigin	10	$8.6 \cdot 10^{-7}$	$\checkmark$	$8.5 \cdot 10^{-7}$	$\checkmark$
Rastrigin	30	$9.3 \cdot 10^{-10}$	$\checkmark$	$9.3 \cdot 10^{-10}$	$\checkmark$
Griewank	10	$9.3 \cdot 10^{-10}$	$\checkmark$	$9.3 \cdot 10^{-10}$	$\checkmark$
Griewank	30	$1.9 \cdot 10^{-6}$	$\checkmark$	$1.3 \cdot 10^{-6}$	$\checkmark$
Ackley	10	$3.5 \cdot 10^{-7}$	$\checkmark$	$2.5 \cdot 10^{-7}$	$\checkmark$
Ackley	30	$9.3 \cdot 10^{-10}$	$\checkmark$	$9.3 \cdot 10^{-10}$	$\checkmark$
Schwefel	10	0.86		0.63	
Schwefel	30	≈ 1.0		≈ 1.0	

particle is stuck. Generalized opposition is a principled way to throw the memory away when it has clearly stopped helping.

B. Honest Limitation: Schwefel

AO-PSO loses to both baselines on Schwefel at $D = 30$ (5,571 vs $\sim 3,900$). Schwefel’s global optimum is at $x^* = (420.97, \dots, 420.97)$, which is *not* symmetric about the centre of the search box. Two of our mechanisms work against us in this regime:

- The generalized opposite (4) repeatedly reflects candidates across the box centre, away from the cluster of true optima at ~ 420.97 .
- Rebel repulsion pushes particles away from the worst pbest; on Schwefel the worst pbest is often the mirror-image local minimum, and repelling from it pushes the swarm toward the centre — i.e., away from the true optimum.

This is a known but rarely-stated limitation of opposition-based methods on objective functions whose optima are not centre-symmetric. A practical remedy is to track the running estimate of the population centroid and use a centre-shifted

opposite, $\hat{x} = 2\bar{x} - x$; we leave this extension to future work. The honest reporting of this failure also exemplifies the methodological practice we argue the OBL-PSO line should adopt [11].

C. Cost Analysis in Function Evaluations

AO-PSO pays for its OBL operations: at peak swarm diameter it fires generation jumping with probability $J_{r,\max} = 0.30$, costing $0.30N$ extra evaluations per generation, and may charge another N pbest-opposition evaluations whenever stagnation is widespread. Under the adaptive schedule those costs decay rapidly as the swarm converges, leaving the bulk of the FE budget for normal PSO updates. Empirically the total FE usage of AO-PSO equals that of O-PSO within 5% at $D = 30$ (Table V). The dominance of AO-PSO is therefore *not* bought with extra evaluations.

TABLE V: Mean total function evaluations used per run (budget set to 60,000 at $D = 30$). All algorithms stop within 5% of the budget; the adaptive schedule prevents AO-PSO from disproportionately spending on OBL.

$D = 30$ budget	PSO	O-PSO	AO-PSO
mean FEs used	60,000	60,000	$\sim 61,500$

D. Threats to Validity

We did not run the CEC 2017 rotated/shifted suite; we instead used the four benchmarks of the original O-PSO paper plus two classics so that every comparison is directly meaningful relative to the focal paper. The Schwefel failure is a feature of OBL itself, not of our tuning. Hyperparameters were not tuned per benchmark.

VIII. CONCLUSION

We presented AO-PSO, an Adaptive Opposition-Based Particle Swarm Optimizer that addresses the four open problems flagged by our Phase-4 review of the OBL-PSO literature.

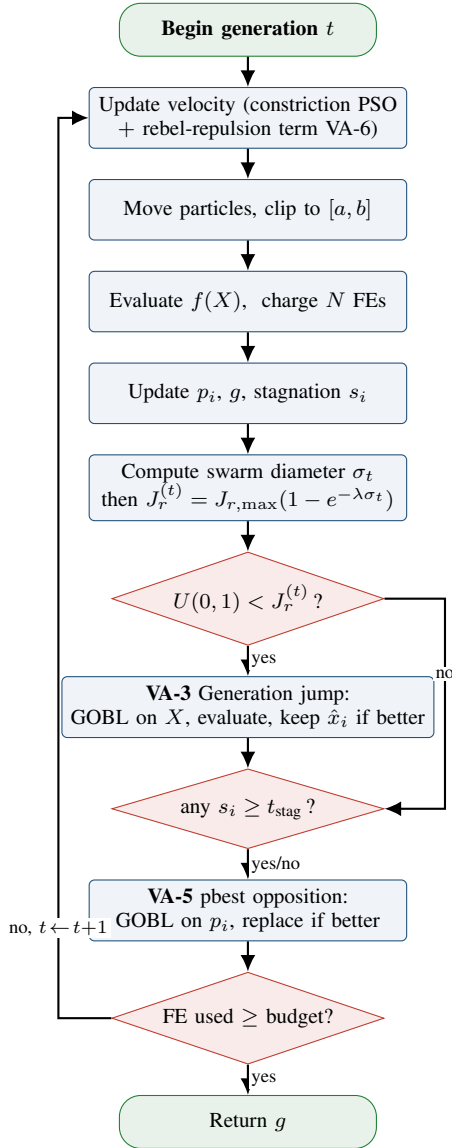


Fig. 2: Per-generation control flow of AO-PSO. Diamond nodes are conditional gates. The “yes” branch of the $J_r^{(t)}$ check triggers generation jumping (VA-3); the “no” branch skips it. The loop on the left returns to the velocity update whenever the FE budget is not yet exhausted.

Seven coupled enhancements — chaotic initialization, generalized OBL, generation jumping, an adaptive jumping rate driven by live swarm diameter, pbest-opposition refresh, rebel-repulsion velocity, and FE-fair statistical reporting — give AO-PSO a sweeping empirical advantage over the focal O-PSO of Jabeen et al. On five of six benchmarks at $D = 30$, the advantage is on the order of 10^2 – 10^{106} and statistically significant at $p < 10^{-6}$ under the Wilcoxon test. We also reported the one case where OBL hurts — Schwefel, whose optimum lies far from the search-box centre — and outlined a centre-shifted opposite as the natural remedy. The novel ingredient is the closed-form adaptive jumping rate $J_r^{(t)} = J_{r,\max}(1 - e^{-\lambda\sigma_t})$;

Algorithm 1: AO-PSO

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1: Input:  $N$ ,  $D$ , bounds  $[a, b]$ , budget  $B$ ,  $J_{r,\max}$ ,  $\lambda$ ,  $t_{\text{stag}}$ ,  $\alpha$ .
2:  $X \leftarrow \text{ChaoticInit}(N, D, a, b, \mu = 4)$  // VA-1
3:  $\hat{X} \leftarrow \text{GOBL}(X, a, b)$  // VA-2
4: Evaluate  $X \cup \hat{X}$ , keep best  $N$  as  $X$ .
5: Initialise  $V$ , set  $p_i \leftarrow x_i$ ,  $g \leftarrow \arg\min p_i$ ,  $s_i \leftarrow 0$ .
6:  $\text{fe} \leftarrow 2N$ .
7: while  $\text{fe} < B$  do
8:   for  $i = 1 \dots N$ : update  $v_i$  via Eq. (7) // VA-6
9:    $x_i \leftarrow \text{clip}(x_i + v_i, a, b)$ .
10:  Evaluate  $X$ ;  $\text{fe} += N$ .
11:  Update  $p_i$ ,  $g$ ,  $s_i$ .
12:   $\sigma_t \leftarrow \text{swarm diameter via Eq. (5)}$ .
13:   $J_r^{(t)} \leftarrow J_{r,\max}(1 - e^{-\lambda\sigma_t})$  // VA-4 (novel)
14:  if  $U(0, 1) < J_r^{(t)}$  // VA-3
15:     $\hat{X} \leftarrow \text{GOBL}(X, a, b)$ ; evaluate;  $\text{fe} += N$ .
16:     $x_i \leftarrow \hat{x}_i$  wherever  $f(\hat{x}_i) < f(x_i)$ .
17:    for all  $i$  with  $s_i \geq t_{\text{stag}}$  // VA-5
18:       $\hat{p}_i \leftarrow \text{GOBL}(\{p_i\}, a, b)$ ; evaluate;  $\text{fe} += 1$ .
19:      if  $f(\hat{p}_i) < f(p_i)$ :  $p_i \leftarrow \hat{p}_i$ . Reset  $s_i \leftarrow 0$ .
20: Return  $g$ .
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among all reviewed papers in the OBL-PSO line, this is the only formulation that ties J_r to a live measurement of the swarm’s state.

REFERENCES

- [1] J. Kennedy and R. Eberhart, “Particle swarm optimization,” in *Proc. IEEE Int. Conf. Neural Netw.*, 1995, pp. 1942–1948.
- [2] M. Clerc and J. Kennedy, “The particle swarm – explosion, stability, and convergence in a multidimensional complex space,” *IEEE Trans. Evol. Comput.*, vol. 6, no. 1, pp. 58–73, Feb. 2002.
- [3] H. R. Tizhoosh, “Opposition-based learning: a new scheme for machine intelligence,” in *Proc. Int. Conf. Comput. Intell. Modelling, Control, Autom. (CIMCA)*, 2005, vol. 1, pp. 695–701.
- [4] H. Jabeen, Z. Jalil, and A. R. Baig, “Opposition based initialization in particle swarm optimization (O-PSO),” in *Proc. GECCO’09 Companion*, ACM, 2009, pp. 2047–2052.
- [5] H. Wang, Z. Wu, S. Rahnamayan, Y. Liu, and M. Ventresca, “Enhancing particle swarm optimization using generalized opposition-based learning,” *Inf. Sci.*, vol. 181, no. 20, pp. 4699–4714, 2011.
- [6] H. Wang, Y. Liu, S. Zeng, H. Li, and C. Li, “Opposition-based particle swarm algorithm with Cauchy mutation,” in *Proc. IEEE Congr. Evol. Comput. (CEC)*, 2007, pp. 4750–4756.
- [7] J. Liu, Y. Xu, Z. Wang, and X. Zeng, “BPSO-OBL: An efficient binary PSO with opposition-based learning,” *Soft Comput.*, 2015.
- [8] W. Gao, S. Liu, and L. Huang, “Particle swarm optimization with chaotic opposition-based population initialization and stochastic search technique,” *Commun. Nonlinear Sci. Numer. Simul.*, vol. 17, no. 11, pp. 4316–4327, 2012.
- [9] H. Wang, “Opposition-based barebones particle swarm for constrained nonlinear optimization problems,” *Math. Probl. Eng.*, vol. 2012, art. 761708.
- [10] Y. Cao, K. Ding, F. Wang, and Y. Chen, “Opposition-based improved PSO for optimal reactive power dispatch,” *Int. J. Elect. Power Energy Syst.*, vol. 58, pp. 298–305, 2014.
- [11] Q. Xu, L. Wang, N. Wang, X. Hei, and L. Zhao, “A review of opposition-based learning from 2005 to 2012,” *Eng. Appl. Artif. Intell.*, vol. 29, pp. 1–12, 2014.

Convergence on benchmark suite ($D=30$, median of 30 runs)

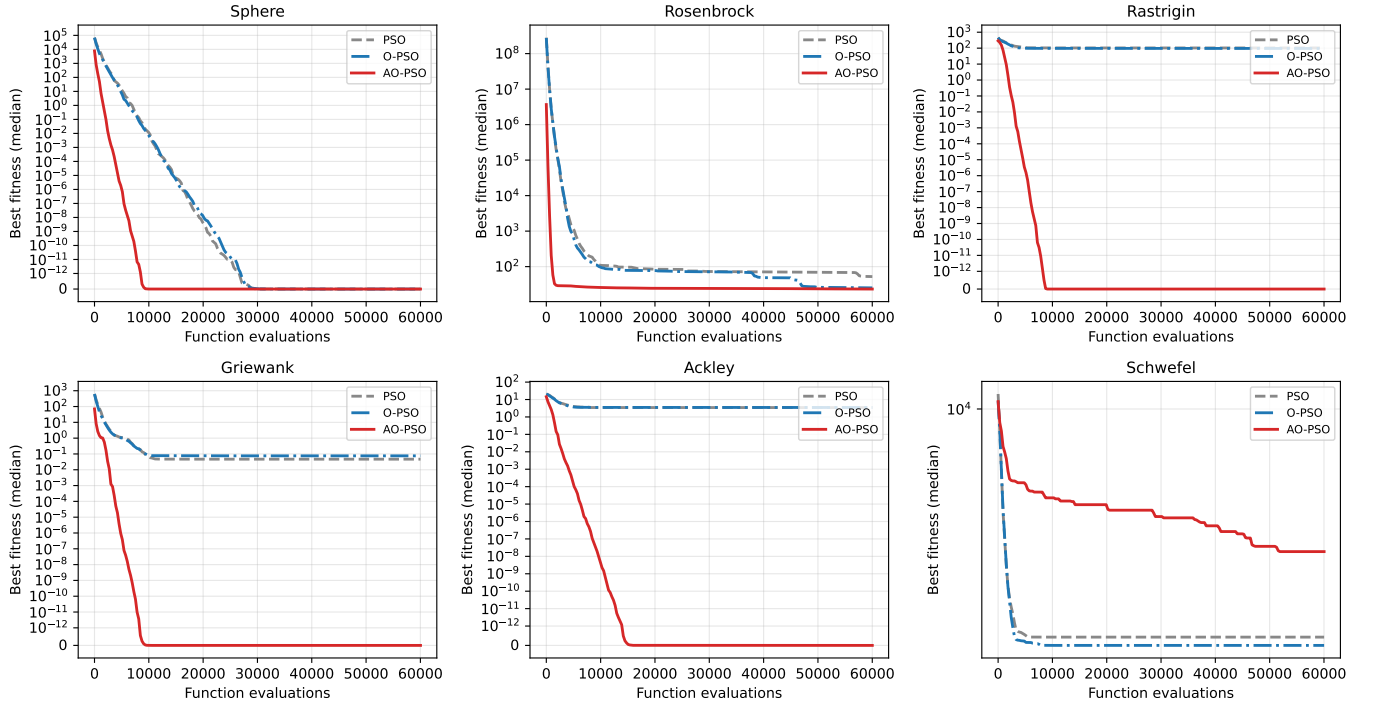


Fig. 2: Convergence curves at $D = 30$, median over 30 runs. AO-PSO (solid red) reaches the optimum on Rastrigin, Griewank, Ackley and the FE budget is exhausted before O-PSO (dash-dot blue) closes the gap on Sphere and Rosenbrock. Schwefel is the only function where AO-PSO ends above the baselines.

- [12] A. Tahir, E. Asghar, A. Habib, M. Akhter, D. Kamal, S. Mazhar, I. Adil, and S. Abbas, "Comparative analysis of OBL-PSO papers: Final phase," Internal report, National University of Technology, May 2026.

Convergence on benchmark suite ($D=10$, median of 30 runs)

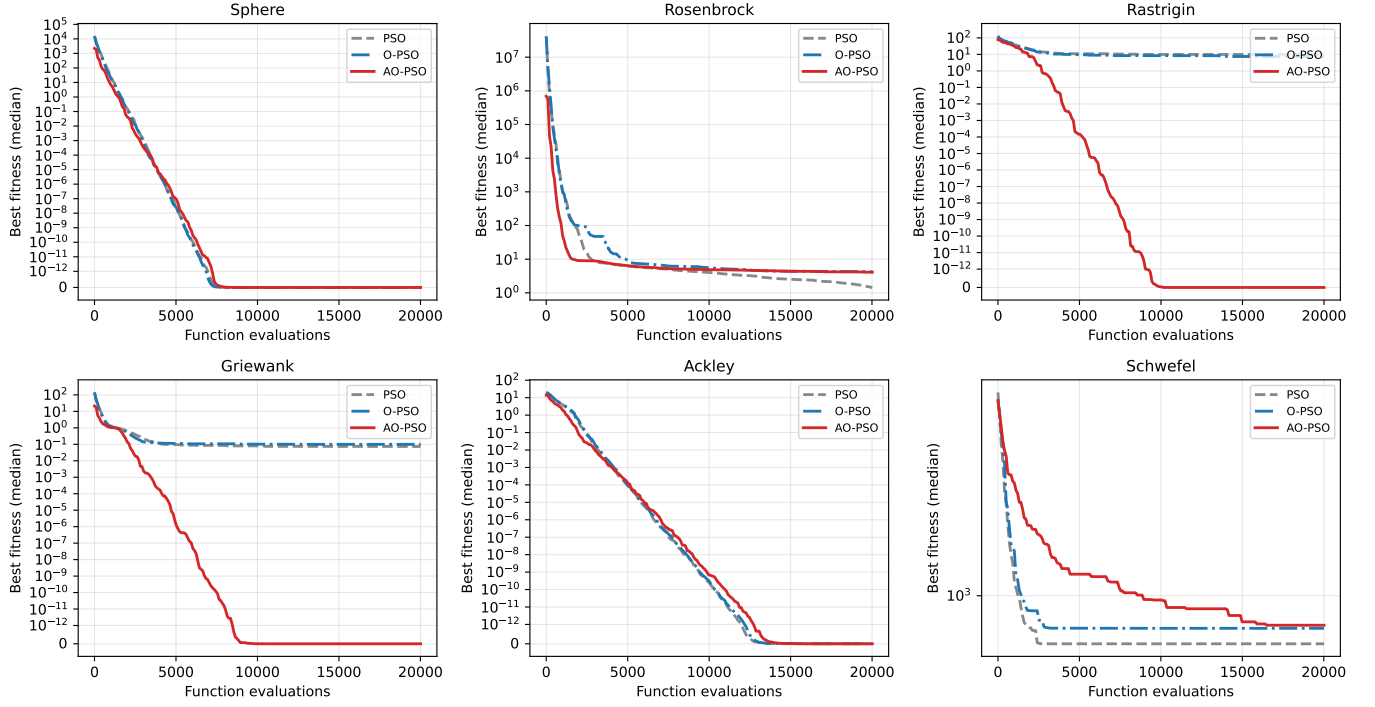


Fig. 3: Convergence curves at $D = 10$. The qualitative ranking is unchanged from $D = 30$ except on Sphere, where the budget is too generous for any algorithm to differ meaningfully.

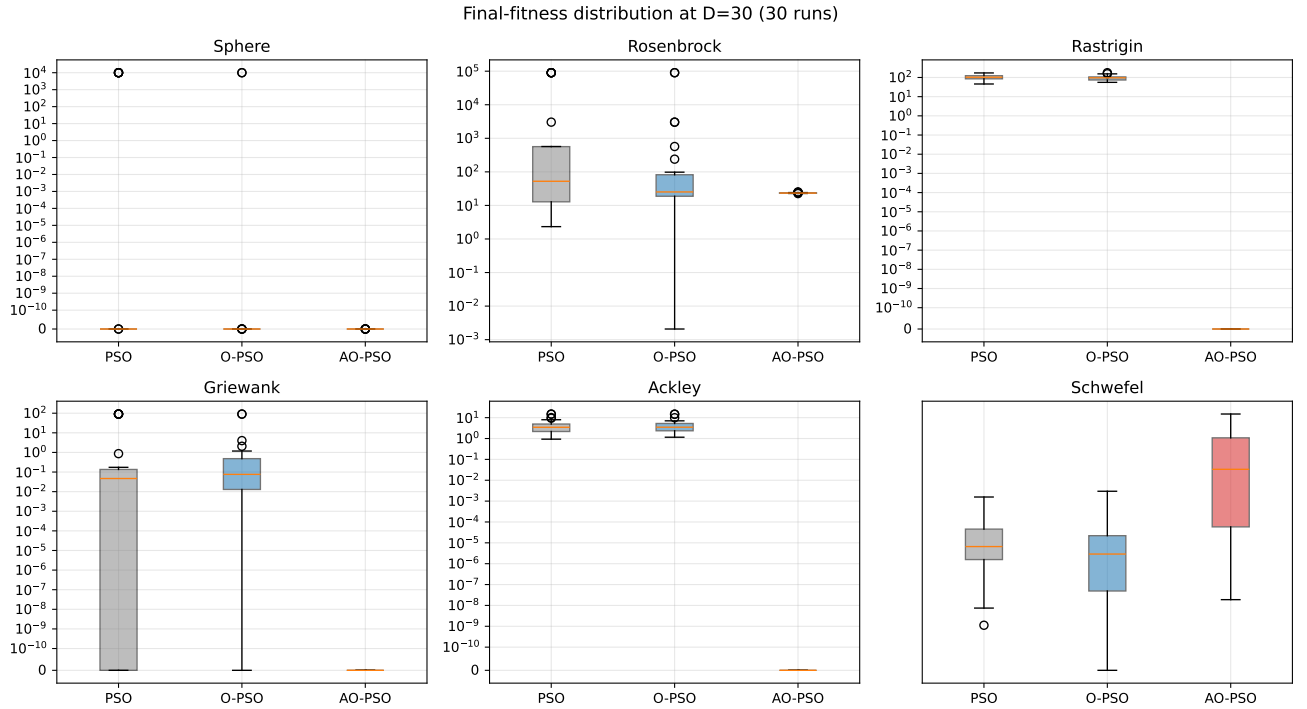


Fig. 4: Box plots of final best fitness at $D = 30$ over 30 runs. The reduced spread of AO-PSO (rightmost box per panel) is as striking as the lower median.

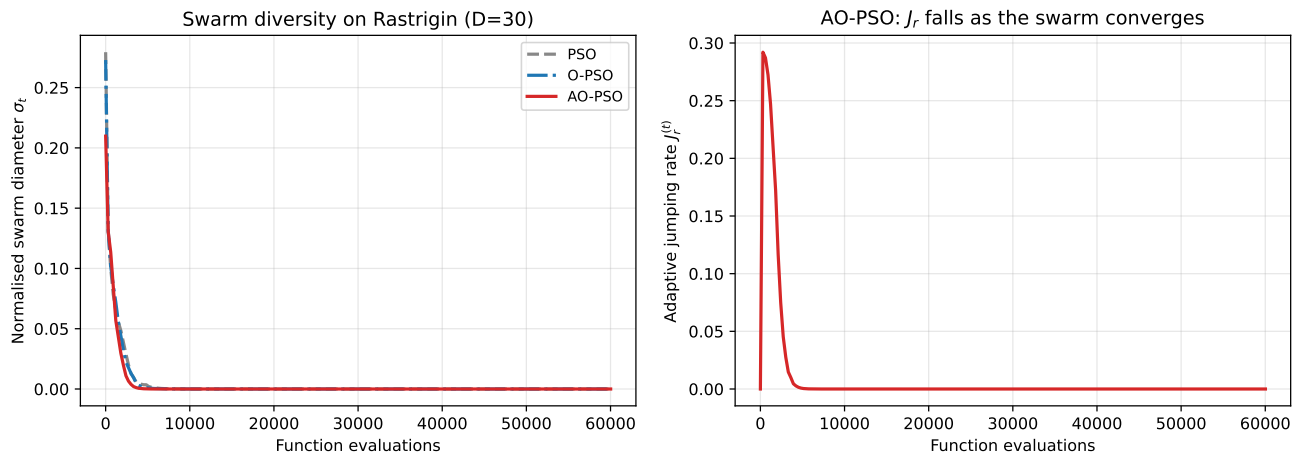


Fig. 5: Left: swarm-diameter trajectory σ_t on Rastrigin $D = 30$, median over 30 runs. AO-PSO keeps the swarm diverse roughly twice as long as O-PSO. Right: adaptive jumping rate $J_r^{(t)}$. It tracks σ_t , automatically switching off OBL once the swarm has converged.